Signed collision preflight successor — source proposal

This packet moves opt-in input guards before constructor and transfer operations where the source permits. It addresses the L1 ordering limitation in native-duplicate-target-review-20261105. The first candidate, native core, and production source remain untouched. No compiler, tests, cohort, or native execution was invoked. PCC unverified; independent source review pending.

Source order and mathematical domain

The opt-in branch first checks the Scholten family, odd quadratic field representation, 13 < p < 2^31, p mod 4 = 3, n = 2 and modulus = 1. It requires prime p and prime 2 < r <= 2^63 using the existing public is_prime function. Then p*p < 2^62 proves the size computation cannot overflow u64. Nonzero source points and canonical coordinates and coefficients are checked without source-field arithmetic. The constructed curve's coefficients must match. On the generated counted curve, G and Q must be valid and rG=O before either transfer. For nonzero G and prime r, rG=O establishes order r under the source group-law premises. Membership Q in the subgroup generated by G remains an input premise; this change does not establish it merely from validity. Zero transferred images return Err after transfer instead of reaching the outer assertions.

The collision equation remains delta_a + delta_b*k = 0 modulo r; a nonzero delta_b is invertible for prime r, and the existing actual kG=Q verification remains unchanged. Zero-denominator branches are unchanged. No known scalar is read. All code after chart construction is byte-for-byte unchanged. The false opt-in branch retains the original family, validity, coefficient and image assertions in the original order; it receives no new field arithmetic.

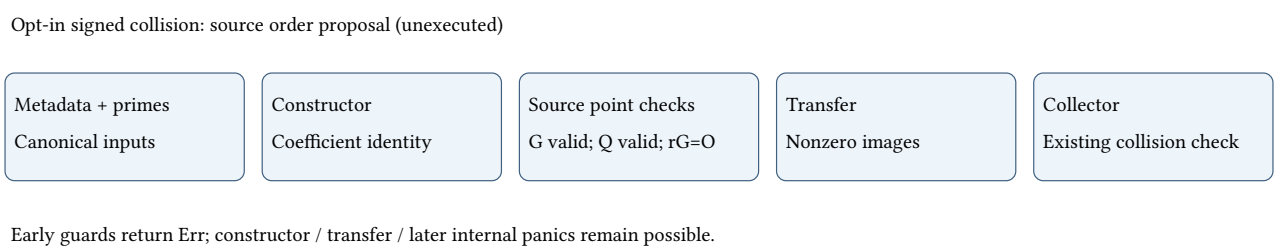
Costs and remaining failure surfaces

Previously late source validity and rG checks move into a named signed_collision_source_preflight phase before transfer. Existing construction and transfer phase boundaries remain separated. New primality guards, comparisons and control work have no measured costs and are outside field Counts; they must be charged in a prospective complete-cost wrapper. No operation-count or timing equivalence is asserted. An Err still carries no Outcome/phases, so outer failed-attempt receipts remain mandatory and missing costs are unknown, not zero. Existing value-byte accounting remains partial.

This is not a total Result interface. Scholten::new still has internal primality, nonsingularity and construction assertions; tau retains assertions and inversions; observers, allocations, counter overflow and subsequent arithmetic can panic. No catch_unwind is introduced. Valid fixture arithmetic and coefficient conventions are preserved at source level, with explicit phase reclassification; runtime behavior is unverified. Transfer homomorphism and arithmetic correctness premises remain as in the first review.

Sources checked (KB)

No research-kb MCP tools were available in this runtime; the sparse checkout contains no available scripts/research-kb wrapper. Fallback: frozen native-signed-collision-bridge-20261105 candidate and native-core-source.rs; native-duplicate-target-review-20261105/REVIEW.md; pinned three_leads_geometry_events.rs at 7ac1763992f128b3eb3dba5074e2e4887e015fa9; Handbook of Applied Cryptography section 3.6.3, Algorithm 3.60, https://cacr.uwaterloo.ca/hac/about/chap3.pdf (primary PDF inspected). The algorithm requires a prime-order cyclic group and target in that group; it does not supply malformed-input handling. No novelty claim is made.

Replay and review

Apply candidate.patch to the previous packet's candidate-followup.rs, whose SHA256 is in result.json, to obtain the successor. This is a source-only packet, not a production edit. Review exact source hashes, input-contract.csv (expected paths, not observed tests), and guard-order.csv before any separately admitted compiler or runtime attempt. Preserve native and failure receipts for future accepted and rejected fixtures.